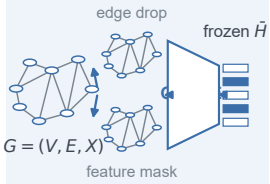
1 Label-free GRACE

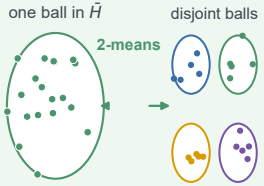


noisy labels

labels not used

only X and A enter GRACE

2 Granular-ball partition



geometry only

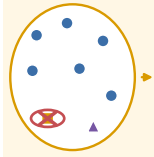
$B_{\max} = 64$

$n_{\min} = 32$

$\delta = 0.05$

no labels

3 Leave-one-out local consensus



$\mathbf{p}_{j,-v}$ unique majority

$\mathbf{q}_{j,-v} = (1-\beta)\mathbf{p}_{j,-v} + \beta\mathbf{e}_m$



tie

$\mathbf{q}_{j,-v} = \mathbf{p}_{j,-v}$

tie-safe

$$\alpha_{j,-v} = \max\left(0, \frac{r_{j,-v} - 1/C}{1 - 1/C}\right)$$

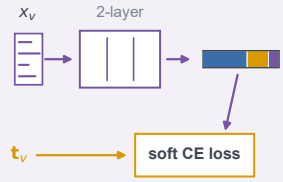
$$\mathbf{t}_v = (1 - \alpha_{j,-v})\mathbf{e}_{\tilde{y}_v} + \alpha_{j,-v}\mathbf{q}_{j,-v}$$

observed label → soft target

soft correction

4 X-only student

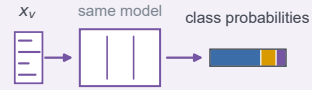
TRAINING



supervision only

DEPLOYMENT

discard A, $\bar{H}$, GRACE and balls



X only · no A · no $\bar{H}$ · no neighbors